Lattice — Feature Showcase

> **Open this file with Lattice** to see every feature rendered live. Switch between **Edit**, **Preview**, and **Dual** view with the toolbar selector. Try `Ctrl/Cmd+Shift+T` to auto-generate the Table of Contents below.

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Text Formatting

Normal paragraph text flows here. Lattice renders Markdown as **close to GitHub** as possible, so files look identical whether viewed in the editor preview, on GitHub, or printed.

Style	Syntax	Result
Bold	`**bold**`	**bold**
Italic	`*italic*`	*italic*
Bold + italic	`***both***`	***both***
Strikethrough	`~~strikethrough~~`	~~strikethrough~~
Inline code	`` `code` ``	`code`
Superscript HTML	X²	X²
Subscript HTML	H₂O	H₂O

Highlight Marks

Highlight marks are a Lattice-exclusive feature. Wrap any text in `==double equals==` to render it with a bright amber background.

This works in **any context** — inside a sentence, ==across *italic* and **bold** spans==, or on its own line:

This entire sentence is highlighted.

> [!NOTE] Highlight marks are **not** rendered inside code blocks or inline code — `==this stays raw==` .

Lists

Unordered

- First item
- Second item
  - Nested item A
  - Nested item B
    - Deeply nested
- Third item with **bold** and *italic* text

Ordered

1. Install Lattice from the releases page
2. Open any `.md` file — double-click or via **File → Open**
3. Choose your view:
   i. **Edit** — raw CodeMirror editor only
   ii. **Preview** — rendered HTML only
   iii. **Dual** — side-by-side (four layout variants)
4. Print with `Ctrl/Cmd+P` — always renders to a clean light theme

Definition-style (using bold + indent)

Local-first All files stay on your machine. No cloud sync required.

Documents as Code Plain-text `.md` files are diff-friendly, version-controlled, and AI-ready.

Task Lists (GFM)

GitHub Flavored Markdown task lists render as interactive checkboxes in Lattice.

- [x] Install Lattice
- [x] Open the demo file
- [x] Explore Edit / Preview / Dual views
- [ ] Try `Ctrl/Cmd+Shift+T` to regenerate the Table of Contents
- [ ] Try `Ctrl/Cmd+Shift+L` to auto-align a pipe table
- [ ] Paste a screenshot with `Ctrl/Cmd+V` — it auto-saves to a sibling folder
- [x] Open a second file in a new window (multi-window support)
- [ ] Try printing to PDF with `Ctrl/Cmd+P`

Tables

Lattice supports **GFM pipe tables** with left, center, and right alignment. Press `Ctrl/Cmd+Shift+L` with the cursor anywhere in the table to auto-format it.

Feature	Edit View	Preview View	Dual View
CodeMirror editor	✓	—	✓
Live rendered preview	—	✓	✓
Scroll sync	—	—	✓
Print (Ctrl+P)	✓	✓	✓
Highlight marks	✓	✓	✓
Mermaid diagrams	—	✓	✓
KaTeX mathematics	—	✓	✓

Code

Inline code

Use backticks for `inline code`. Great for `file/paths`, `commands`, and `variable_names`.

Fenced code blocks

Lattice passes code fences to the preview unchanged — syntax highlighting is provided by the renderer.

```
// Lattice highlight-mark plugin — converts ==text== to <mark> elements
export const MARK_RE = /==((?:[^=\n]|=(?!=))+?)==/g;

export function splitAtMarks(text: string): HastNode[] {
  const parts: HastNode[] = [];
  let lastIndex = 0;
  MARK_RE.lastIndex = 0;
  let match: RegExpExecArray | null;
  while ((match = MARK_RE.exec(text)) !== null) {
    if (match.index > lastIndex)
      parts.push({ type: 'text', value: text.slice(lastIndex, match.index) });
```

```
    parts.push({
      type: 'element', tagName: 'mark',
      properties: {},
      children: [{ type: 'text', value: match[1] }],
    });
    lastIndex = match.index + match[0].length;
  }
  if (lastIndex < text.length)
    parts.push({ type: 'text', value: text.slice(lastIndex) });
  return parts;
}
```

```
// Tauri command — read file with conflict detection
#[tauri::command]
pub async fn read_file(path: String) -> Result<FileReadResult, String> {
    let content = fs::read_to_string(&path)
        .map_err(|e| format!("Cannot read {path}: {e}"))?;
    let hash = compute_sha256(&content);
    Ok(FileReadResult { content, hash })
}
```

```
# Build a production release
export LATTICEBUILD_NO="MyBuild-001"
npm run tauri build
```

Blockquotes

> **Single-level blockquote.** Great for callouts, citations, or side notes.

> Nested blockquotes are also supported:
>
> > Inner quote — "*Documents as code, not documents as data.*"
> >
> > — Lattice design philosophy

GitHub-style Alerts (rendered by GitHub and Lattice)

> [!NOTE] Useful information that users should know, even when skimming.

> [!TIP] Optional advice for a better experience.

> [!IMPORTANT] Key information users need to know to achieve their goal.

> [!WARNING] Urgent information that needs immediate attention.

> [!CAUTION] Advises about risks or negative outcomes.

Mathematics — KaTeX

Lattice renders mathematics via **KaTeX**. Use `$...$` for inline and `$$...$$` for display math.

Inline math

Einstein's famous equation $E = mc^2$ and the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ render inline.

The Euler identity $e^{i\pi} + 1 = 0$ is often called the most beautiful equation in mathematics.

Display math — Calculus

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$$

$$\frac{d}{dx}\left[\int_{a}^{g(x)} f(t)\,dt\right] = f(g(x)) \cdot g'(x)$$

Display math — Linear Algebra

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x} \quad \Longrightarrow \quad \det(\mathbf{A} - \lambda\mathbf{I}) = 0$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Display math — Series & Limits

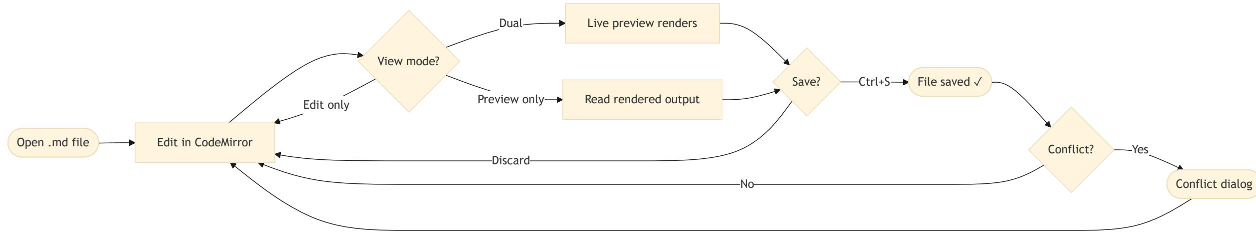
$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\lim_{n \to \infty}\left(1 + \frac{1}{n}\right)^n = e$$

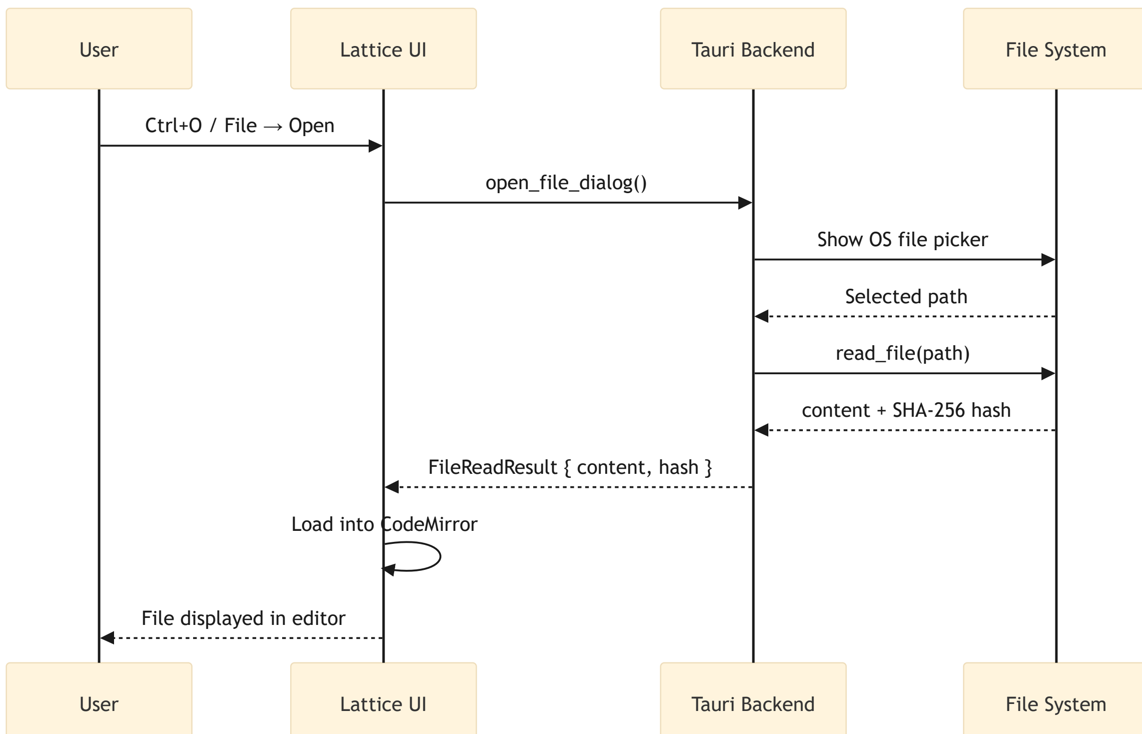
Mermaid Diagrams

Lattice renders ```` ```mermaid ``` ```` fenced blocks inline in the preview.

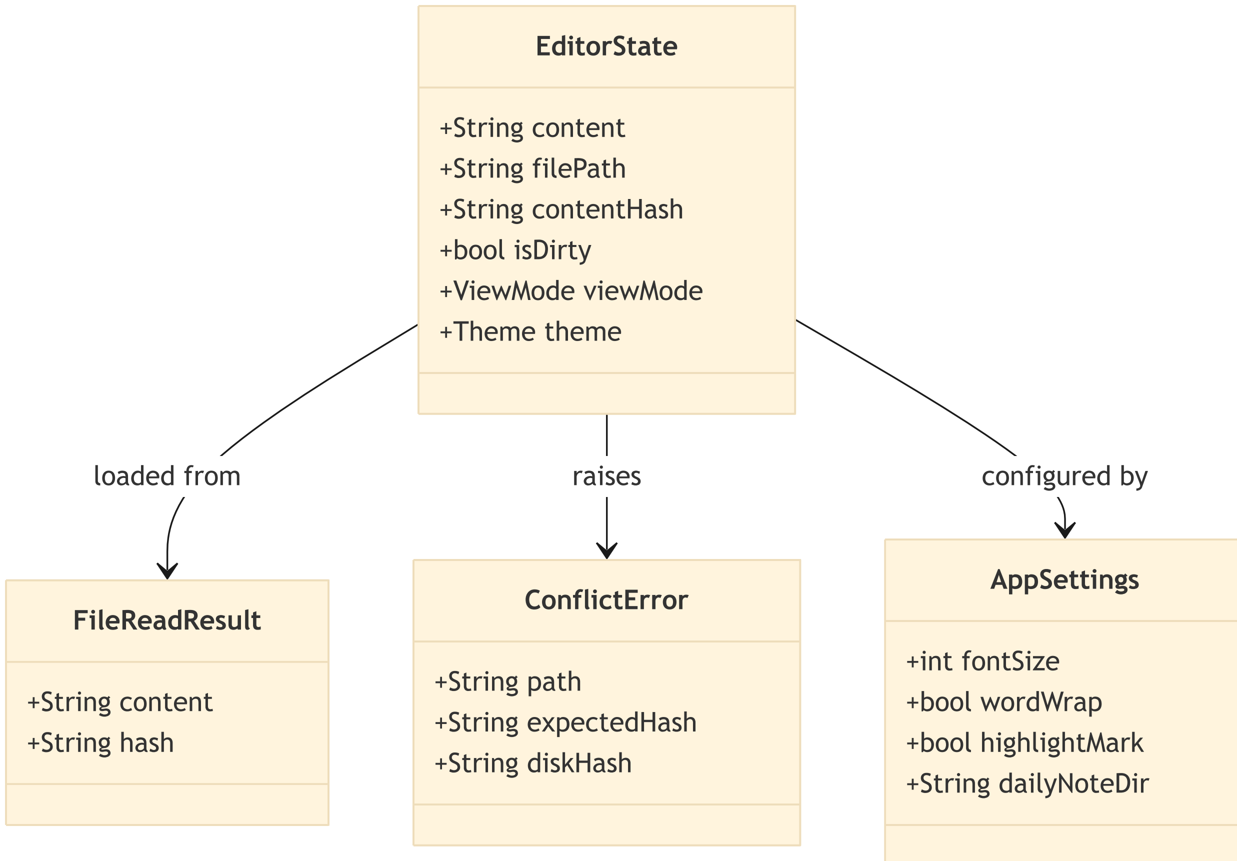
Flowchart — Lattice Edit Cycle



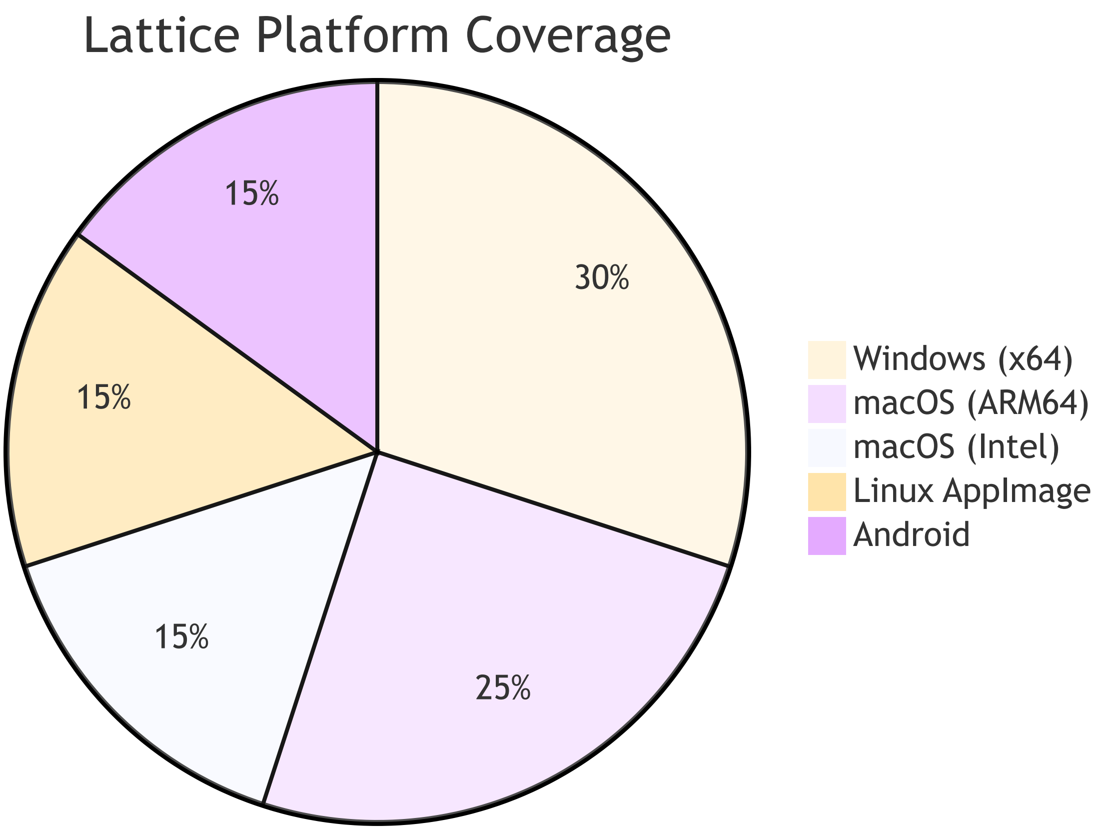
Sequence Diagram — File Open Flow



Class Diagram — Core Data Model



Pie Chart — Supported Platforms



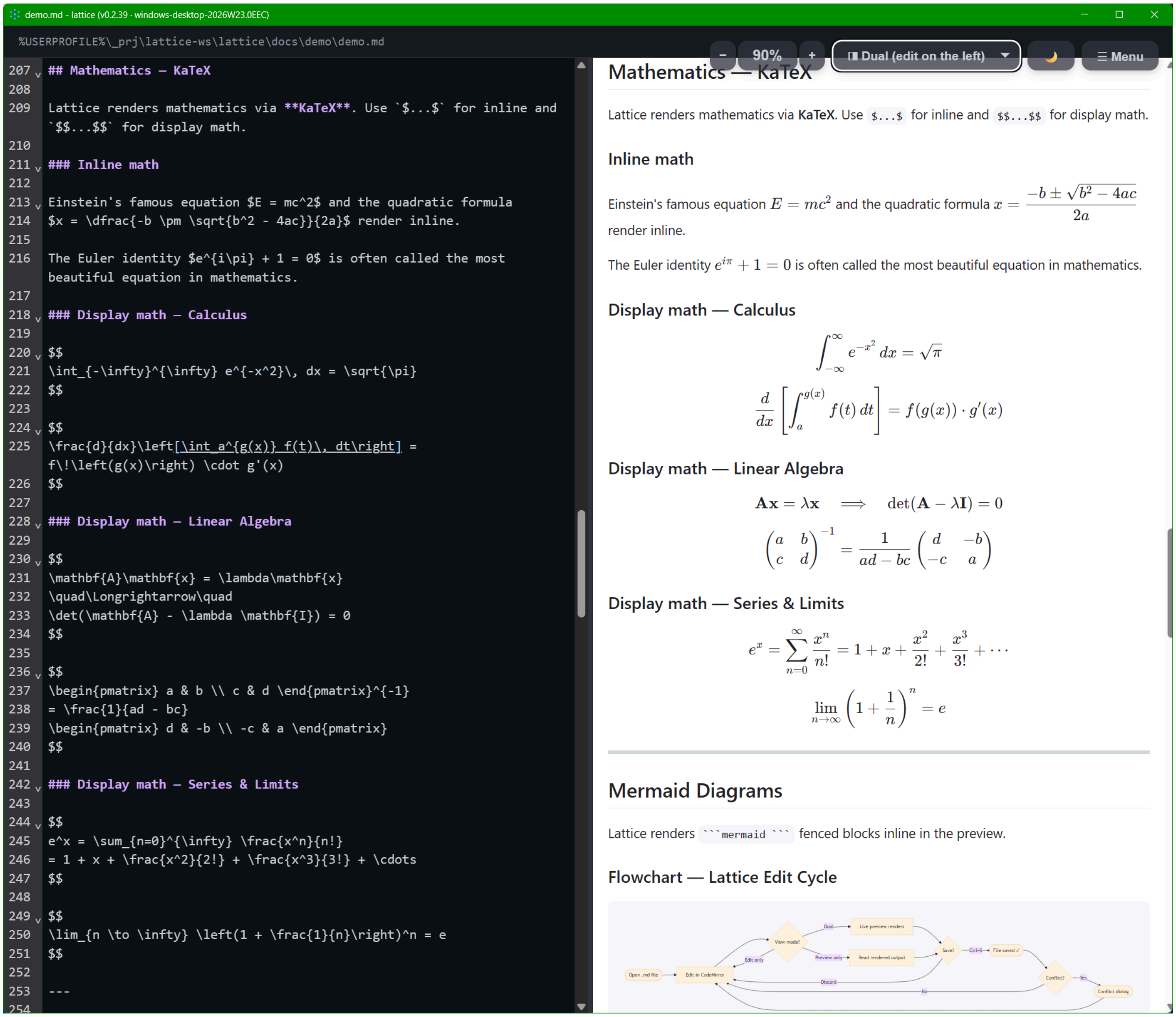
Images

Paste images directly into Lattice with `Ctrl/Cmd+V`.
The image file is automatically saved to a sibling folder named after the current document, and a Markdown reference is inserted at the cursor position:

```
![Image](demo_assets/img_1780574419628.png)

![Image](demo_assets/img_1780575215039.png)
```

Here is how it renders in `Dual (edit on the left)` mode:



Here is how it renders in `Dual (edit on the bottom)` mode:

demo.md - lattice (v0.2.39 · windows-desktop-2026W23.0EEC)

%USERPROFILE%_prj\lattice-ws\lattice\docs\demo\demo.md

\- 70% + Dual (edit on bottom) ☰ Menu

Mathematics — KaTeX

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Display math — Calculus

$$
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$$

$$
\frac{d}{dx}\left[\int_a^{g(x)} f(t)\, dt\right] = f\!\left(g(x)\right) \cdot g'(x)
$$

Display math — Linear Algebra

$$
\mathbf{A}\mathbf{x} = \lambda\mathbf{x} \quad\Longrightarrow\quad \det(\mathbf{A} - \lambda\mathbf{I}) = 0
$$

$$
\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}
$$

Display math — Series & Limits

$$
e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots
$$

```
---

## Mathematics — KaTeX

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### Display math — Calculus

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### Display math — Linear Algebra

$$
\mathbf{A}\mathbf{x} = \lambda\mathbf{x}
\quad\Longrightarrow\quad
```

```
\det(\mathbf{A} - \lambda \mathbf{I}) = 0
```

> **Tip:** The assets folder name is derived from the document filename, keeping related files grouped together and easy to version-control.

Links

- Lattice on GitHub — source code and issue tracker
- Releases Page — download installers
- Release Site — GitHub Pages release portal
- Relative link: System Architecture Demo
- Relative link: Physics Equations Demo

Auto-link: https://github.com/blessia-blessini/lattice

HTML in Markdown

Lattice passes safe inline HTML through to the preview. Useful for fine-grained formatting unavailable in pure Markdown:

<details> <summary>Click to expand — collapsible section</summary>
Hidden content revealed on click. Great for long appendices or spoilers.

```json
{
  "product": "Lattice",
  "version": "0.3.0",
  "license": "AGPL-3.0"
}
```

</details>

HTML element	Use case
`<mark>`	Rendered by `==highlight==` syntax
`<details>`	Collapsible sections
`<sup>`	Superscripts: X²
`<sub>`	Subscripts: CO₂

HTML element	Use case
` `	Explicit line break
`<kbd>`	Keyboard keys: <kbd>Ctrl</kbd>+<kbd>S</kbd>

Keyboard Shortcuts Reference

Action	Windows / Linux	macOS
Save	<kbd>Ctrl+S</kbd>	<kbd>Cmd+S</kbd>
Open file	<kbd>Ctrl+O</kbd>	<kbd>Cmd+O</kbd>
New window	<kbd>Ctrl+N</kbd>	<kbd>Cmd+N</kbd>
Generate / refresh TOC	<kbd>Ctrl+Shift+T</kbd>	<kbd>Cmd+Shift+T</kbd>
Auto-align table	<kbd>Ctrl+Shift+L</kbd>	<kbd>Cmd+Shift+L</kbd>
Paste image from clipboard	<kbd>Ctrl+V</kbd>	<kbd>Cmd+V</kbd>
Print / Save as PDF	<kbd>Ctrl+P</kbd>	<kbd>Cmd+P</kbd>
Increase font size	<kbd>Ctrl++</kbd>	<kbd>Cmd++</kbd>
Decrease font size	<kbd>Ctrl+-</kbd>	<kbd>Cmd+-</kbd>